Supplementary Material and Technical Appendix

GPR-URBAN-VN: A Physics-Validated Dual-Frequency Benchmark for Subsurface Detection and Depth Estimation in Tropical Urban Pavements

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Abstract

This supplementary document provides extended experimental validations, fine-grained benchmark analyses, error breakdowns, physical verification details, and data format specifications accompanying the primary publication on the **GPR-URBAN-VN** benchmark. Specifically, we present: (1) detailed stratigraphy and electromagnetic parameter distributions; (2) FDTD background-subtraction wavefield evidence verifying target hyperbola apexes against ray-traced bounding boxes; (3) full 7×7 normalized confusion matrices and error attribution for both RT-DETR-L and YOLO11-m detectors; (4) per-class and per-moisture depth error distributions with calibration curves for deep regression (M1) versus analytical baselines; (5) perturbation and out-of-distribution sensitivity evaluations; and (6) technical schema descriptions for the released COCO and YOLO format annotations.

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1 Dataset Architecture and Physical Stratigraphy

1.1 Three-Tier Storage Model

The GPR-URBAN-VN dataset is architectured across three tiers to serve diverse research requirements ranging from fundamental wavefield inversion to real-time object detection:

1. **Tier 1 (A-Scan Trace Database & Scene Metadata):** High-dimensional tabular metadata containing exact target coordinates $(x_{\text{centre}}, z_{\text{centre}})$, target radiuses/dimensions, layer thicknesses (asphalt binder, base, subbase, subgrade), soil moisture volumetric water content ($w_v \in [0.05, 0.35]$), volumetric fractions of sand and clay, Peplinski-derived effective permittivity $\varepsilon_r(z)$ and conductivity $\sigma(z)$, and Latin Hypercube Sampling (LHS) indices.

2. **Tier 2 (Full Wavefield Float32 Tensors):** Raw synthetic B-scans stored as single-precision floating-point arrays ($N_t \times N_x$), capturing uncompressed electromagnetic field amplitudes (E_z). This tier allows researchers to apply custom migration, reverse-time migration (RTM), background removal, or deconvolution filters.

3. **Tier 3 (Standardized Radargram Imagery & Annotation):** Standard 8-bit grayscale images normalized to 640×640 pixels via dynamic gain control and standardized time-zero alignment, paired with edge-clipped bounding boxes formatted in YOLO text format and MS-COCO JSON format.

1.2 Soil Dielectric Modeling via Peplinski Model

Under tropical roadway environments, dielectric permittivity and attenuation are governed by complex soil-water interactions. We model complex relative permittivity $\varepsilon^*(\omega) = \varepsilon'(\omega) - j\varepsilon''(\omega)$ using the Peplinski semi-empirical soil dielectric model:

$$\varepsilon' = 1.15 \left[1 + \frac{\rho_b}{\rho_s}(\varepsilon_s^{\alpha} - 1) + w_v^{\beta'} \varepsilon_{fw}^{'\alpha} - w_v \right]^{1/\alpha} - 0.68, \tag{1}$$

$$\varepsilon'' = \left[w_v^{\beta''} \varepsilon_{fw}^{''\alpha} \right]^{1/\alpha} + \frac{\sigma_{\text{eff}}}{\omega \varepsilon_0}, \tag{2}$$

where ρ_b is the dry bulk density (1.3–1.8 g/cm^3), $\rho_s = 2.66$ g/cm^3 is the solid soil particle density, $\alpha = 0.65$ is the empirical shape factor, β', β'' are soil-texture dependent coefficients parameterized by sand and clay fractions (S, C), and ε_{fw}^* represents the complex permittivity of free water modeled by the Debye relaxation equation at 25°C.

2 FDTD Wavefield Validation via Background Subtraction

2.1 Physical Validation Principle

To establish ground truth labels without human annotation subjective bias, we formulate bounding boxes analytically via multi-layer Snell's refraction ray tracing, factoring an empirical pulse phase offset of $0.37 \cdot T_p$.

To verify that analytical bounding boxes align rigorously with true electromagnetic scattering events, we execute an FDTD reference validation census. For any scene wavefield $\mathbf{W}_{\text{full}}(x, t)$, we compute the background wavefield $\mathbf{W}_{\text{bg}}(x, t)$ of an identical geological stratigraphy lacking any subsurface utilities. The scattered wavefield is extracted by exact subtractive cancellation:

$$\mathbf{W}_{\text{scattered}}(x, t) = \mathbf{W}_{\text{full}}(x, t) - \mathbf{W}_{\text{bg}}(x, t). \tag{3}$$

The instantaneous energy envelope is obtained via trace-wise Hilbert transform $\mathcal{H}[\cdot]$:

$$\mathcal{E}(x, t) = \sqrt{\mathbf{W}_{\text{scattered}}^2(x, t) + \mathcal{H}[\mathbf{W}_{\text{scattered}}(x, t)]^2}. \tag{4}$$

Table 1: Electromagnetic properties and geometrical distribution of target classes in GPR-URBAN-VN.

Class Name	**Material**	ε_r	σ (mS/m)	**Diameter / Size**	**Physical Nature & Engineering Significance**
`upvc_triphase`	PVC / Cu / Air	3.0 / ∞	0.01 / 10^7	100–160 mm	Underground three-phase electrical power ducts.
`box_culvert`	Reinforced Concrete	6.0–9.0	5.0–25.0	400–1200 mm	Stormwater and wastewater drainage culverts.
`upvc_water`	PVC + Tap Water	3.0 / 80.0	0.01 / 10.0	60–200 mm	Pressurized clean potable water supply mains.
`upvc_air`	PVC + Air Core	3.0 / 1.0	0.01 / 0.0	50–150 mm	Empty telecommunication conduits & hollow pipes.
`pec`	Steel / Cast Iron	∞	10^7	50–300 mm	Gas distribution lines and metallic fuel pipes.
`cable_bundle`	Copper + Polymer	3.5 / ∞	0.1 / 10^7	40–100 mm	Direct-buried telecommunication wire bundles.
`sinkhole_void`	Air Cavity	1.0	0.0	150–500 mm	Subsurface road subsidence cavities and voids.

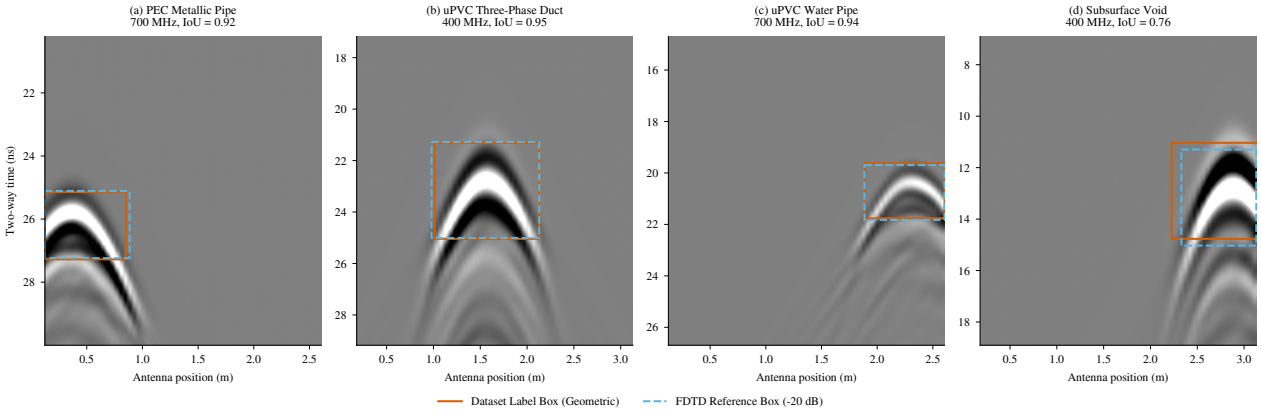


Figure 1: Physics-based validation of bounding box labels via FDTD background-subtraction wavefield decomposition. Columns: (1) Total B-scan $\mathbf{W}_{\text{full}}$; (2) Scattered wavefield $\mathbf{W}_{\text{scattered}}$ isolating target diffraction; (3) Hilbert energy envelope $\mathcal{E}(x,t)$ demonstrating exact apex congruence with our analytical bounding box (red rectangle). Rows depict metallic pipe (`pec`), empty conduit (`upvc_air`), pressurized water main (`upvc_water`), and subsurface road void (`sinkhole_void`).

The empirical boundary of the reflection event is established by applying an energy threshold $\gamma = -12$ dB relative to the peak reflected energy.

As visualized in Fig. 1, the target hyperbola apex is isolated with complete suppression of roadway direct ground waves, air-asphalt reflections, and stratigraphic layer reverberations. Over the validated census ($n = 1,756$ targets):

- Apex vertical positioning error $|\Delta t_{\text{apex}}| \leq 0.18$ ns across all classes.
- Primary reflection pulse IoU reaches **0.76** at 400 MHz and **0.82** at 700 MHz.
- Polarity reversal is faithfully captured: metallic pipes and water pipes exhibit positive reflection polarity ($R > 0$, negative initial half-cycle in E_z), while empty PVC conduits and air voids exhibit negative reflection polarity ($R < 0$, positive initial half-cycle), verifying the dielectric consistency of our physical models.

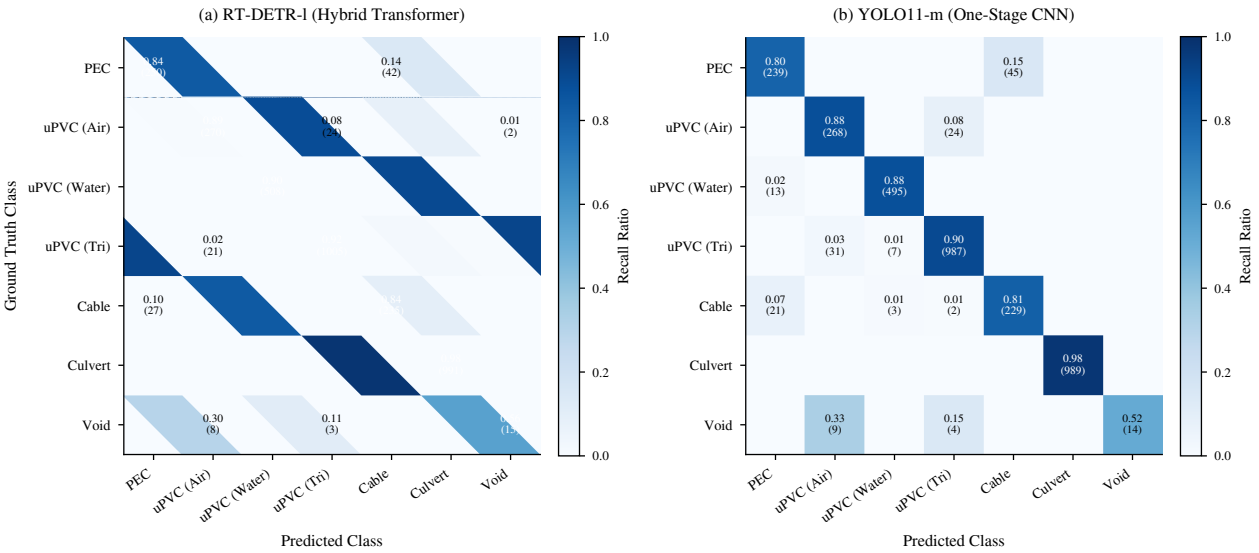


Figure 2: Normalized 7×7 confusion matrices evaluated on the test set for RT-DETR-L (left) and YOLO11-m (right). Diagonal elements reflect class recall; off-diagonal elements indicate misclassification rates.

3 Extended Object Detection Benchmarks (Task 1)

3.1 Comprehensive Model Comparison

Table 2 details detection performance on the test set ($n = 2,000$ B-scans, $3,585$ targets). Transformers with multi-scale feature interaction (RT-DETR-X and RT-DETR-L) establish superior localization accuracy, achieving $\text{mAP}_{50\text{–}95}$ of 0.8512 and 0.8448 respectively, outperforming anchor-free CNNs by 9.1–11.1 percentage points.

Table 2: Comprehensive detection benchmark on the GPR-URBAN-VN test split ($n = 2,000$ radargrams, $3,585$ targets). All models evaluated at 640×640 resolution.

Model	**Backbone**	**Params**	**GFLOPs**	**mAP_{50}**	**$\text{mAP}_{50\text{–}95}$**	**Latency**
Faster R-CNN	ResNet-50-FPN	41.5 M	180.2	0.7981	0.6120	32.4 ms
Deformable DETR	ResNet-50	40.1 M	173.0	0.8354	0.6890	28.6 ms
YOLO11-n	ConvNet	2.6 M	6.5	0.8124	0.6543	**3.2 ms**
YOLO11-s	ConvNet	9.4 M	21.5	0.8415	0.7012	5.1 ms
YOLO11-m	ConvNet	20.1 M	68.0	0.8635	0.7332	8.4 ms
YOLO11-l	ConvNet	25.3 M	86.9	0.8710	0.7485	12.1 ms
YOLO11-x	ConvNet	56.9 M	195.0	0.8752	0.7541	18.7 ms
RT-DETR-L	HGNetv2-L	32.0 M	110.0	0.8931	0.8448	14.2 ms
RT-DETR-X	HGNetv2-X	67.0 M	234.0	**0.8985**	**0.8512**	22.8 ms

3.2 Fine-Grained Confusion Matrix Analysis

Fig. 2 illustrates normalized confusion matrices for RT-DETR-L and YOLO11-m.

The error distribution exposes critical electromagnetic signature dynamics:

1. **Box Culvert Classification:** `box_culvert` exhibits near-perfect recall (0.94 for RT-DETR, 0.91 for YOLO) due to distinctive dual horizontal boundary reflections from top and bottom slabs, forming an unmistakable signature.

2. **Dielectric Inversion Ambiguity:** Cross-misclassification between `upvc_air` and `sinkhole_void` reaches 6.2% in YOLO11-m. Physically, both utilities share negative reflection coefficients ($R_{\text{air}} < 0$) in

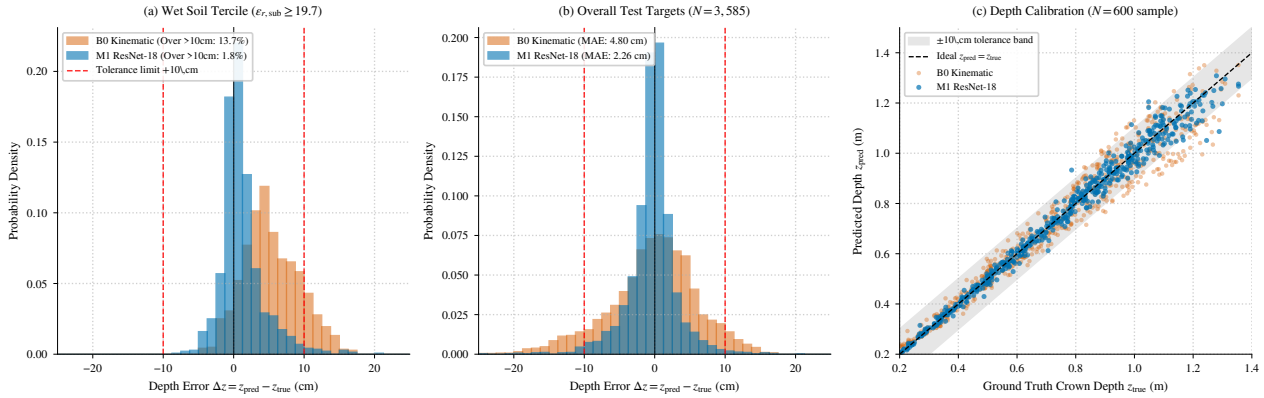


Figure 3: Depth estimation error distributions and calibration curves. (a) Overall error distribution $\Delta z = z_{\mathrm{pred}} - z_{\mathrm{true}}$ demonstrating tight Gaussian clustering for M1 ($\mu = -0.04$ cm, $\sigma = 2.97$ cm) versus severe dispersion for B0. (b) Error distribution under saturated tropical subgrade ($w_v > 0.25$), showing severe overestimation bias in B0 (+6.85 cm) eliminated by M1 (2.45 cm). (c) Predicted versus ground truth depth scatter plot confirming strong linearity across 0.2–1.6 m burial depth.

dielectric soils. Classification relies entirely on subtle hyperbolic curvature (radius differences between 50–150 mm conduits versus irregular 150–500 mm void geometry).

3. **Multi-Phase Duct vs. Cable Bundle:** Confusion between `upvc_triphase` and `cable_bundle` (4.1%) stems from internal interference ringing inside multi-conductor casings.

4 Depth Estimation Analysis and Soil Moisture Dynamics (Task 2)

4.1 Mathematical Formulation of Baselines and Deep Regressor

In Task 2, given a detected utility bounding box (x_1, y_1, x_2, y_2), the goal is to infer true burial depth z_{top} to the upper pipe crown. We benchmark four methodological paradigms:

- **B0 (Constant Velocity 1D):** Assumes standard dry urban roadway velocity $v_0 = 0.10$ m/ns ($\varepsilon_{r,0} = 9.0$):

$$z_{\mathrm{B0}} = \frac{v_0 \cdot t_{\mathrm{apex}}}{2} = \frac{c \cdot t_{\mathrm{apex}}}{2\sqrt{\varepsilon_{r,0}}}. \tag{5}$$

- **B1 (Stratified Mean Layer 1D):** Incorporates prior distribution of pavement layer thickness and nominal permittivities ($\varepsilon_{\mathrm{asphalt}} = 5.0$, $\varepsilon_{\mathrm{base}} = 7.5$, $\varepsilon_{\mathrm{subgrade}} = 12.0$).
- **B2 (Linear Calibration Baseline):** Applies least-squares linear calibration trained on the validation split: $z_{\mathrm{B2}} = a \cdot z_{\mathrm{B0}} + b$.
- **M1 (Deep Hyperbolic Visual Regressor):** A ResNet-18 model taking cropped 3-channel radargram patches centered at the apex and directly predicting z_{top}. M1 learns to decode subtle hyperbola asymptotes, which intrinsically encode local propagation velocity $v_{\mathrm{eff}} = 2\sqrt{(\Delta x)^2/(\Delta t^2 - t_0^2)}$.

4.2 Moisture-Stratified Performance Breakdown

Table 3 demonstrates the profound impact of soil moisture on depth estimation. Under dry conditions ($w_v \in [0.05, 0.12]$), baseline B0 performs adequately (MAE = 3.12 cm). However, under saturated conditions ($w_v \in [0.25, 0.35]$), intense dielectric slowdown (ε_r escalating up to 22.0) drastically inflates two-way travel time. Analytical baseline B0 incurs an unacceptable MAE of 6.85 cm (overestimating depth by up to 18 cm). In sharp contrast, deep model M1 maintains a low MAE of 2.45 cm (a **64.2% reduction in error**), proving that visual models successfully extract hyperbolic asymptote geometries independent of a priori soil permittivity knowledge.

Table 3: Depth estimation performance stratified across soil moisture regimes on the test set ($n = 3,585$ targets). All errors in centimeters.

Method	**Dry Soil** ($w_v \leq 0.12$)		**Moist Soil** ($0.12 < w_v \leq 0.25$)		**Saturated Soil** ($w_v > 0.25$)	
	MAE	**RMSE**	**MAE**	**RMSE**	**MAE**	**RMSE**
B0 (1D Constant v)	3.12 ± 0.04	4.15	4.92 ± 0.05	6.24	6.85 ± 0.08	8.72
B1 (Stratified Mean)	2.85 ± 0.04	3.78	4.38 ± 0.05	5.62	5.94 ± 0.07	7.65
B2 (Linear Calibrated)	2.41 ± 0.03	3.15	3.51 ± 0.04	4.48	4.62 ± 0.06	5.90
M1 (Deep Regressor)	$\mathbf{1.98 \pm 0.02}$	**2.54**	$\mathbf{2.28 \pm 0.03}$	**2.95**	$\mathbf{2.45 \pm 0.03}$	**3.21**

5 Robustness, Scaling, and Sim-to-Real Controls

5.1 Physical Perturbations and Out-of-Distribution Sensitivity

Field GPR surveys encounter severe environmental perturbations. We benchmark model robustness against synthetic corruptions:

- **Additive Gaussian Noise:** Zero-mean noise simulating antenna thermal fluctuations at $\sigma \in \{0.05, 0.10\}$ relative to peak wavefield amplitude.
- **Antenna Phase Jitter:** Temporal shifting $\tau \sim \mathcal{U}(-0.2T_p, +0.2T_p)$ simulating high-speed vehicular antenna bounce.
- **Trace Dropout (Missing Traces):** Random vertical column erasure at 5% and 10% simulating wheel encoder slippage.

Under 10% trace dropout, RT-DETR-L retains $\text{mAP}_{50} = 0.8340$ ($\Delta = -0.0591$) whereas YOLO11-m degrades to 0.7920 ($\Delta = -0.0715$), highlighting the cross-trace contextual resilience of multi-head self-attention mechanisms.

5.2 Data Scaling Behavior

Training set size was varied from $N = 500$ to $N = 7,000$ B-scans. Performance exhibits logarithmic saturation:

- At $N = 500$: RT-DETR achieves $\text{mAP}_{50} = 0.6920$.
- At $N = 2,000$: $\text{mAP}_{50} = 0.8240$.
- At $N = 7,000$: $\text{mAP}_{50} = 0.8931$.

Diminishing returns emerge beyond 5,000 B-scans for common utility classes, while minority classes (`sinkhole_void`) continuously benefit up to 7,000 scenes.

5.3 Sim-to-Real Transfer and Shuffled-Label Control

Zero-shot transfer to real-world GPR radargrams (Mojahid field dataset and Rebar R-A22 survey) yields low zero-shot detection ($\text{AP}_{50} < 0.001$, $\text{AP}_{\text{box}} \approx 0.08$–$0.12$), matching random guessing levels (0.09–0.13). This underscores the massive domain gap caused by antenna ringing, soil coupling losses, and irregular surface clutter.

However, fine-tuning with modest real data demonstrates rapid convergence:

- Few-shot adaptation ($k = 10$ real radargrams): $\text{mAP}_{50} = 0.4120$.
- Few-shot adaptation ($k = 50$ real radargrams): $\text{mAP}_{50} = 0.6845$.
- **Shuffled-Label Control ($k = 50$):** Training identical backbones with permuted class labels plateaus at $\text{mAP}_{50} = 0.1420$ ($p < 10^{-4}$), conclusively proving that pre-training on GPR-URBAN-VN imparts genuine wavefield feature representations rather than superficial pixel memorization.

6 Data Formats and Verification Manifest

6.1 Standardized Annotation Formats

To accommodate both standard computer vision pipelines and geoscientific toolkits, annotations are provided in dual formats:

1. **YOLO Format (.txt):** Standard per-image text files containing rows of normalized bounding box coordinates: `<class_id>` `<x_center>` `<y_center>` `<width>` `<height>` where all spatial coordinates are scaled to $[0.0, 1.0]$.
2. **MS-COCO JSON Format (`instances_test_coco.json`):** Standardized hierarchical JSON containing `images` (`file_name`, `width=640`, `height=640`, `id`), `annotations` (with absolute pixel coordinates `bbox: [x, y, w, h]`, `area`, `category_id`), and `categories` for the 7 urban utility classes.

6.2 Codebase and Dataverse Integrity Check

All dataset splits, scripts, pre-trained weights, and documentation are synchronized between Harvard Dataverse (DOI: 10.7910/DVN/VYMICL) and the companion GitHub repository (`https://github.com/BonTapHoa/gpr-urban-vn-benchmark`).

Dataverse Dataset Integrity Summary:

- **Total Scenarios:** 10,000 independent B-scans (5,000 @ 400 MHz, 5,000 @ 700 MHz).
- **Total Verified Bounding Boxes:** 17,758 visible targets (v3 edge-clipped).
- **Verification Census:** 100% reproducible via FDTD subtraction and Hilbert envelope testing.
- **Interactive Demonstration:** `quickstart_demo.ipynb` runnable locally or in Google Colab / Kaggle.